

PRACTICE 1

```
mysql> CREATE TABLE bank_transactions (  
  ->     txn_id INT PRIMARY KEY,  
  ->     customer_name VARCHAR(50),  
  ->     branch_name VARCHAR(50),  
  ->     transaction_type VARCHAR(20),  
  ->     amount DECIMAL(10,2),  
  ->     transaction_date DATE  
  -> );  
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> ALTER TABLE bank_transactions  
  -> ADD account_no VARCHAR(20);  
Query OK, 0 rows affected (0.14 sec)  
Records: 0  Duplicates: 0  Warnings: 0
```

```
mysql> ALTER TABLE bank_transactions  
  -> MODIFY customer_name VARCHAR(100);  
Query OK, 0 rows affected (0.16 sec)  
Records: 0  Duplicates: 0  Warnings: 0
```

```
mysql> ALTER TABLE bank_transactions  
  -> RENAME TO customer_transactions;  
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> TRUNCATE TABLE customer_transactions;  
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> DROP TABLE customer_transactions;  
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> -- Step 1: Select your database
mysql> USE bank_db;
Database changed
```

```
mysql> -- Step 2: Verify it is selected (should show 'bank_db' instead of NULL)
mysql> SELECT database();
+-----+
| database() |
+-----+
| bank_db    |
+-----+
1 row in set (0.00 sec)
```

```
mysql> -- Step 2: Insert your data
mysql> INSERT INTO bank_transactions VALUES
  -> (101,'Ravi','Hyderabad','Deposit',5000,'2024-01-05'),
  -> (102,'Sita','Hyderabad','Withdrawal',2000,'2024-01-06'),
  -> (103,'Kiran','Vijayawada','Deposit',12000,'2024-01-08'),
  -> (104,'Anil','Vizag','Deposit',8000,'2024-01-10'),
  -> (105,'Priya','Hyderabad','Withdrawal',3500,'2024-01-11'),
  -> (106,'Ramesh','Vizag','Deposit',15000,'2024-01-12'),
  -> (107,'Keerthi','Vijayawada','Withdrawal',1000,'2024-01-13'),
  -> (108,'Rahul','Hyderabad','Deposit',9000,'2024-01-14'),
  -> (109,'Sneha','Vizag','Withdrawal',4000,'2024-01-15'),
  -> (110,'Madhu','Vijayawada','Deposit',11000,'2024-01-16');
Query OK, 10 rows affected (0.01 sec)
Records: 10  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO bank_transactions
  -> VALUES
  -> (111,'Venu','Vizag','Deposit',7000,'2024-01-18');
Query OK, 1 row affected (0.01 sec)
```

```
mysql> INSERT INTO bank_transactions
  -> VALUES
  -> (111,'Venu','Vizag','Deposit',7000,'2024-01-18');
Query OK, 1 row affected (0.01 sec)
```

```
mysql> UPDATE bank_transactions
  -> SET amount = 5000
  -> WHERE id = 105;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 1  Changed: 0  Warnings: 0
```

```
mysql> DELETE FROM bank_transactions
  -> WHERE id = 111;
Query OK, 2 rows affected (0.01 sec)
```

```
mysql> SELECT * FROM bank_transactions;
```

id	name	city	type	amount	date
101	Ravi	Hyderabad	Deposit	5000	2024-01-05
102	Sita	Hyderabad	Withdrawal	2000	2024-01-06
103	Kiran	Vijayawada	Deposit	12000	2024-01-08
104	Anil	Vizag	Deposit	8000	2024-01-10
105	Priya	Hyderabad	Withdrawal	5000	2024-01-11
106	Ramesh	Vizag	Deposit	15000	2024-01-12
107	Keerthi	Vijayawada	Withdrawal	1000	2024-01-13
108	Rahul	Hyderabad	Deposit	9000	2024-01-14
109	Sneha	Vizag	Withdrawal	4000	2024-01-15
110	Madhu	Vijayawada	Deposit	11000	2024-01-16

```
10 rows in set (0.00 sec)
```

```
mysql> SELECT *
-> FROM bank_transactions
-> WHERE type = 'Deposit';
```

id	name	city	type	amount	date
101	Ravi	Hyderabad	Deposit	5000	2024-01-05
103	Kiran	Vijayawada	Deposit	12000	2024-01-08
104	Anil	Vizag	Deposit	8000	2024-01-10
106	Ramesh	Vizag	Deposit	15000	2024-01-12
108	Rahul	Hyderabad	Deposit	9000	2024-01-14
110	Madhu	Vijayawada	Deposit	11000	2024-01-16

```
6 rows in set (0.00 sec)
```

```
mysql> SELECT *
-> FROM bank_transactions
-> ORDER BY amount DESC;
```

id	name	city	type	amount	date
106	Ramesh	Vizag	Deposit	15000	2024-01-12
103	Kiran	Vijayawada	Deposit	12000	2024-01-08
110	Madhu	Vijayawada	Deposit	11000	2024-01-16
108	Rahul	Hyderabad	Deposit	9000	2024-01-14
104	Anil	Vizag	Deposit	8000	2024-01-10
101	Ravi	Hyderabad	Deposit	5000	2024-01-05
105	Priya	Hyderabad	Withdrawal	5000	2024-01-11
109	Sneha	Vizag	Withdrawal	4000	2024-01-15
102	Sita	Hyderabad	Withdrawal	2000	2024-01-06
107	Keerthi	Vijayawada	Withdrawal	1000	2024-01-13

```
10 rows in set (0.00 sec)
```

SECTION 3 : DCL (DATA CONTROL LANGUAGE)

```
mysql> CREATE USER 'BranchManager'@'localhost'  
-> IDENTIFIED BY 'manager123';  
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> GRANT ALL PRIVILEGES  
-> ON bank_db_practice.customer_transactions  
-> TO 'BranchManager'@'localhost';  
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> REVOKE SELECT  
-> ON bank_db_practice.customer_transactions  
-> FROM 'Auditor1'@'localhost';  
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> |
```

```
mysql> REVOKE ALL PRIVILEGES  
-> ON bank_db_practice.customer_transactions  
-> FROM 'BranchManager'@'localhost';  
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> SHOW GRANTS FOR 'Auditor1'@'localhost';  
+-----+  
| Grants for Auditor1@localhost |  
+-----+  
| GRANT USAGE ON *.* TO `Auditor1`@`localhost` |  
+-----+  
1 row in set (0.00 sec)
```

SECTION 4 : TCL (MySQL)

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE customer_transactions
    -> SET amount = 6000
    -> WHERE txn_id = 101;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0  Changed: 0  Warnings: 0
```

```
mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE customer_transactions
    -> SET amount = 7000
    -> WHERE txn_id = 101;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0  Changed: 0  Warnings: 0
```

```
mysql>
mysql> SAVEPOINT SP1;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql>
mysql> UPDATE customer_transactions
    -> SET amount = 9000
    -> WHERE txn_id = 102;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0  Changed: 0  Warnings: 0
```

```
mysql>
mysql> ROLLBACK TO SP1;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)
```

